

Final Project Report

Interpretable Sentiment Analysis of Customer Reviews

Using Deep Learning

IE 7615: Deep Learning for AI

Wiktoria Lasek

GitHub: https://github.com/Wiktoria0110/IE7615_Project_Deep_Learning

1. Problem Setting

Millions of customer reviews are posted online every day across platforms such as Amazon and Yelp. For businesses, these reviews represent an invaluable source of feedback about their products and services. However, manually reading and analysing thousands of reviews is impossible at scale. A small business owner with 500 reviews, a product manager tracking a new launch, or a customer service team monitoring complaints - all face the same challenge: understanding what customers truly think, quickly and accurately.

Traditional NLP approaches such as bag-of-words models and rule-based classifiers can label reviews as positive or negative, but they fail to capture context, sarcasm, and nuanced language. More critically, they cannot explain WHY a review was classified a certain way. A business does not just need to know that a review is negative - it needs to know whether customers are complaining about delivery, product quality, or customer service. Deep learning offers both higher accuracy and interpretability through attention mechanisms - making it the ideal approach for this problem.

2. Problem Definition

This project aims to build an automated system that reads customer reviews and predicts whether the experience was positive or negative. Using a progressive deep learning pipeline - from simple baseline to state-of-the-art transformer - the goal is to identify which model best captures customer sentiment from text, and to visually explain which words and phrases drove each prediction.

The primary contribution of this project is a comparative study of four architectures – Logistic Regression, Feedforward Neural Network, LSTM, and DistilBERT - with an interpretability layer via attention visualization and cross-domain robustness analysis on Yelp reviews.

The outcome is a practical tool that businesses can use to automatically monitor customer feedback at scale, identify recurring issues, and make data-driven decisions to improve their products and services.

3. Data Sources

This project uses two publicly available datasets from Kaggle:

Amazon Customer Reviews Dataset [1]: A large-scale dataset of Amazon product reviews with binary sentiment labels. Available at:

<https://www.kaggle.com/datasets/bittlingmayer/amazonreviews>

Yelp Reviews Sentiment Dataset [2]: A dataset of Yelp restaurant and business reviews with binary sentiment labels, used exclusively for cross-domain generalization testing.

Available at: <https://www.kaggle.com/datasets/thedevastator/yelp-reviews-sentiment-dataset>

Both datasets are freely available, contain no personally identifiable information, and raise no ethical concerns.

4. Data Description

Table below provides a detailed description of all three dataset splits used in this project.

Attribute	Amazon Train	Amazon Test	Yelp (corss-domain)
Total rows	3,600,000	400,000	38,000
Sample used	200,000	40,000	5,000
Number of columns	2	2	2
Column names	Label, review	Label, review	Text, label
Label format	_label_1/2 -> 0/1	_label_1/2 -> 0/1	0/1
Class balance	50/50	50/50	50/50
Missing values	0	0	0
Mean review length	78 words	78 words	133 words
Median review length	70 words	70 words	97 words

Purpose	Train all models	Standard evaluation	Cross-domain test
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5. Data Exploration

5.1 Missing Value Analysis

A missing value check was performed across all datasets. Zero missing values were found in all columns across all three datasets. This means no imputation was required and all sampled reviews could be used for modelling directly.

5.2 Label Distribution

Both Amazon train and test datasets are perfectly balanced - 1,800,000 positive and 1,800,000 negative reviews in the full training set (50/50). After sampling, the balance was maintained: 100,020 positive and 99,980 negative in the 200,000 training sample. The Yelp cross-domain dataset is also balanced - 2,540 negative and 2,460 positive in the 5,000 sample. This balanced distribution is ideal for binary classification and means F1-score and accuracy provide similar information.

5.3 Review Length Analysis

Review length was analysed to guide preprocessing and model design decisions. Table below summarizes the key statistics.

Statistic	Amazon	Yelp
Mean length	78.48 words	133.37 words
Median length	70 words	97 words
Std deviation	42.86 words	121.65 words
Min length	8 words	1 word
Max length	218 words	988 words

Yelp reviews are notably longer than Amazon reviews - mean 133 words vs 78 words. This difference in writing style makes Yelp a more challenging and meaningful cross-

domain test. The max sequence length of 128 tokens used for LSTM and DistilBERT covers the median Amazon review length (70 words) effectively.

6. Deep Learning Tasks

6.1 Label Conversion

Amazon labels were converted from _label_1/_label_2 format to binary 0/1: _label_1 -> 0 (negative), _label_2 -> 1 (positive). This conversion was applied to both train and test samples before any modelling.

6.2 Text Preprocessing

Different preprocessing strategies were applied depending on the model - a key design decision justified by the different input requirements of each architecture:

Logistic Regression and FNN (heavy preprocessing): Text was converted to lowercase, punctuation and special characters were removed using regular expressions (re.sub), and stop words were removed using NLTK's English stop word list (~179 words). This produces clean, noise-free text suitable for TF-IDF vectorization.

LSTM (minimal preprocessing): Text was tokenized and padded to max length 128 using Keras Tokenizer (num_words=10,000). Stop words were NOT removed as LSTM benefits from context words like 'not' which change meaning.

DistilBERT (no preprocessing): Raw text was passed directly to the HuggingFace pipeline. DistilBERT uses its own subword tokenizer and benefits from original casing and punctuation.

6.3 Feature Engineering — TF-IDF

TF-IDF (Term Frequency-Inverse Document Frequency) vectorization converted clean text into numerical feature vectors for LR and FNN. TF-IDF assigns each word a weight based on how frequently it appears in a review relative to how common it is across all reviews (Salton & Buckley, 1988). Two configurations were used: max_features=10,000 for LR (fitted on 200,000 reviews) and max_features=5,000 for FNN (fitted on 50,000

reviews due to memory constraints). The vectorizer was fitted on training data only to prevent data leakage.

6.4 Sequence Preparation for LSTM

For LSTM, reviews were tokenized into integer sequences using Keras Tokenizer (num_words=10,000) and padded to a fixed length of 128 tokens using pad_sequences. This fixed-length representation is required for batch processing in the LSTM model.

7. Deep Learning Models and Methods

7.1 Logistic Regression (Baseline)

Logistic Regression serves as the traditional ML baseline. It learns a weight for each TF-IDF feature through supervised training - assigning positive weights to sentiment-positive words and negative weights to sentiment-negative words. Trained with max_iter=1000 and random_state=42. In addition to classification, LR coefficients provide interpretability - the top 10 positive and negative words are extracted and compared against DistilBERT attention scores.

7.2 Feedforward Neural Network

A two-layer feedforward neural network was implemented using PyTorch. Architecture: Linear(5000->256) ->ReLU ->Dropout(0.3) -> Linear(256->128) -> ReLU ->Dropout(0.3) ->Linear(128->2). Adam optimizer (lr=0.001), CrossEntropyLoss, 5 epochs, batch size 256. Dropout regularization prevents overfitting. Trained on 50,000 reviews due to memory constraints when converting sparse TF-IDF matrix to dense tensors.

7.3 LSTM (Long Short-Term Memory)

LSTM [3] reads text sequentially and maintains a memory cell that captures long-range dependencies. Keras Sequential model: Embedding(10000, 128, input_length=128) ->LSTM(64) ->Dropout(0.3) ->Dense(1, sigmoid). Adam optimizer, binary crossentropy loss, 3 epochs, batch size 256, validation split 0.1. Trained on 200,000 Amazon reviews.

7.4 DistilBERT (Transformer)

DistilBERT [4] is a pre-trained transformer that uses self-attention to capture complex linguistic patterns. The pre-trained pipeline distilbert-base-uncased-finetuned-sst-2-english was loaded via HuggingFace Transformers - already fine-tuned on the SST-2 sentiment dataset. Used without additional fine-tuning on Amazon data. Max sequence length 128. Attention weight visualization generates heatmaps showing which tokens most influenced each prediction - compared against LR coefficients for interpretability analysis [5].

8. Performance Evaluation

All models were evaluated using the following metrics:

Accuracy: The proportion of correctly classified reviews. Simple but reliable for balanced datasets like Amazon (50/50).

F1-Score (Primary Metric): The harmonic mean of precision and recall. Designated as primary metric as it provides a balanced measure of model performance.

Precision: Of all reviews predicted as positive/negative, how many actually were?

Recall: Of all actual positive/negative reviews, how many were correctly identified?

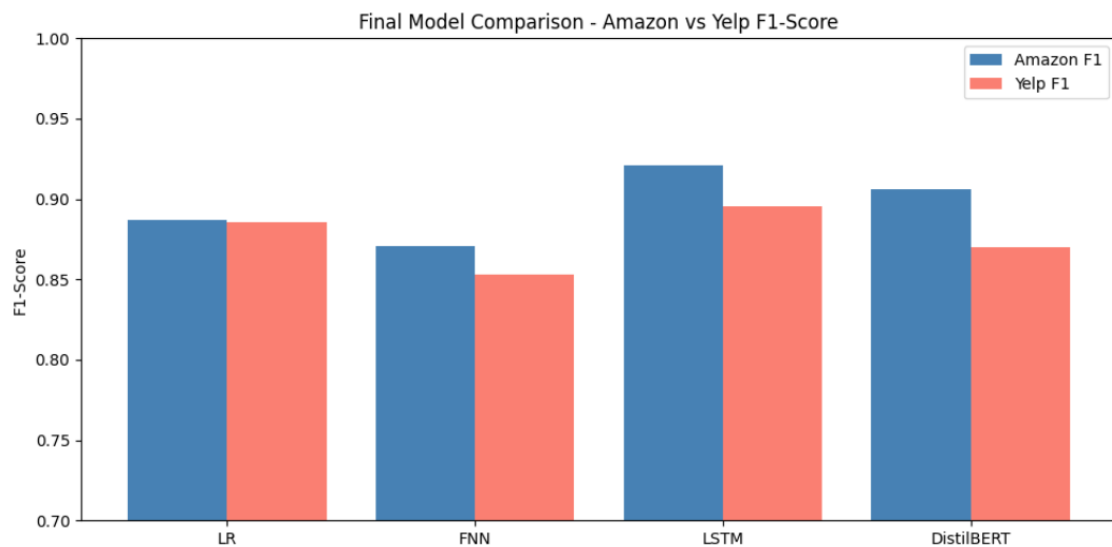
Cross-Domain F1: F1-score on Yelp reviews - measures generalization to a different review platform and writing style.

Domain Drop: The difference in F1 between Amazon test and Yelp - measures how much performance degrades on the cross-domain test.

9. Project Results

9.1 Model Performance Comparison

Model	Amazon Acc	Amazon F1	Yelp Acc	Yelp F1	Drop
Logistic Regression	88.81%	88.85%	87.84%	88.24%	-0.61%
FNN	87.54%	87.68%	84.74%	85.55%	-2.13%
LSTM	91.50%	92.10%	89.30%	89.58%	-2.52%
DistilBERT	90.60%	90.58%	87%	87.03%	-3.55%



9.2 Key Findings

LSTM is the best model - achieving 92.10% F1 on Amazon and 89.58% on Yelp. Its ability to read text sequentially and remember context gives it a clear advantage over bag-of-words approaches. The training loss decreased consistently from 0.2871 to 0.1591 across 3 epochs, with validation accuracy reaching 92.03%.

LSTM outperforms DistilBERT because DistilBERT was used as a pre-trained pipeline without fine-tuning on Amazon data. The pipeline was fine-tuned on movie reviews (SST-2) which differ from product reviews in vocabulary and style. Direct fine-tuning on Amazon data would likely close this gap.

FNN underperforms LR - trained on only 50,000 reviews with 5,000 TF-IDF features due to memory constraints, compared to LR which used 200,000 reviews and 10,000 features. This confirms that larger training samples improve neural network performance.

All models generalize well to Yelp - LR shows the smallest drop (-0.61%), confirming that simple word patterns transfer well across domains. All models demonstrate genuine sentiment learning beyond Amazon-specific vocabulary.

9.3 Interpretability Results

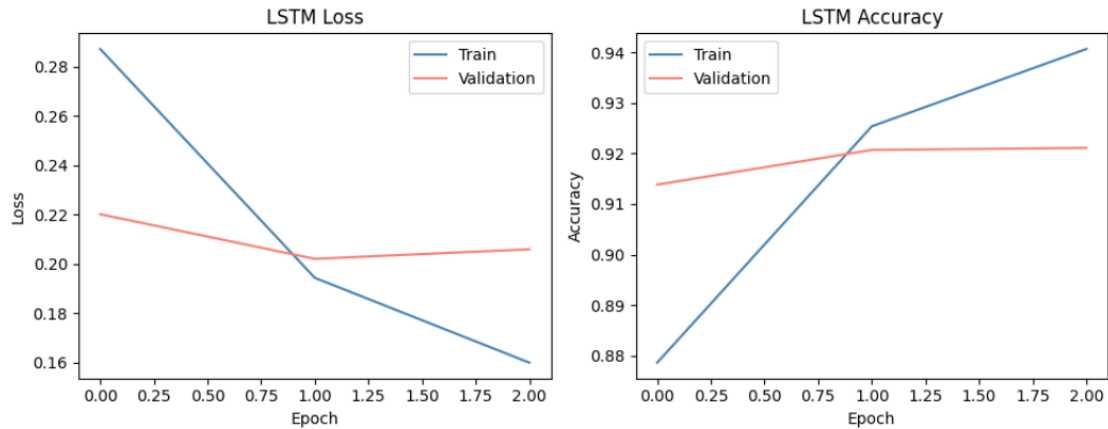
LR coefficients reveal the most sentiment-driving words:

Top positive words	Top negative words
Great, excellent, perfect, awesome, best, amazing, love, highly, wonderful	Worst, disappointing, waste, disappointment, disappointed, boring, poor, terrible

DistilBERT attention heatmaps complement LR coefficients by showing contextual attention - for positive reviews words like 'amazing' and 'best' receive highest attention, while for negative reviews 'terrible' and 'waste' dominate. This interpretability comparison confirms both models learned meaningful sentiment patterns.

9.4 LSTM Training History

Epoch	Train loss	Train acc	Val loss	Val acc
1	0.2871	87.86%	0.2201	91.39%
2	0.1933	92.47%	0.2068	91.88%
3	0.1591	93.92%	0.2165	92.03%



10. Impact of Project Outcomes

The deep learning pipeline developed in this project has significant practical value for businesses and researchers.

Automated review monitoring: The LSTM model can automatically classify thousands of customer reviews in seconds - enabling real-time monitoring of customer sentiment at scale across any product category.

Explainable AI: The combination of LR coefficients and DistilBERT attention heatmaps provides two complementary interpretability approaches - making AI decisions transparent and trustworthy. Businesses can understand exactly which words drive positive or negative classifications.

Cross-domain deployment: All models generalize well to Yelp reviews, demonstrating that the pipeline can be deployed across different review platforms without retraining.

Progressive architecture insight: The comparative study provides empirical evidence for when deep learning adds value over traditional methods - LSTM's +3.25% improvement over LR demonstrates that sequential modeling is beneficial for sentiment analysis.

Limitations and future work: The main limitation is that DistilBERT was not fine-tuned on Amazon data. Future work should include direct fine-tuning of DistilBERT, testing larger

transformer models such as RoBERTa, expanding cross-domain testing to additional platforms, and deploying the best model as a real-time review classification API.

Generative AI Usage Statement

Generative AI tools (Claude) were used only for: brainstorming project ideas, researching model architectures and relevant references, and occasional guidance on code structure and syntax. No text, code, figures, or analysis was directly copied without modification and understanding.

References

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